

CE-BENCH: TOWARDS A RELIABLE CONTRASTIVE EVALUATION BENCHMARK OF INTERPRETABILITY OF SPARSE AUTOENCODERS

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ABSTRACT

Probing with sparse autoencoders is a promising approach for uncovering interpretable features in large language models (LLMs). However, the lack of automated evaluation methods has hindered their broader adoption and development. In this work, we introduce CE-Bench, a novel and lightweight contrastive evaluation benchmark for sparse autoencoders, built on a curated dataset of contrastive story pairs. We conduct comprehensive ablation studies to validate the effectiveness of our approach. Our results show that CE-Bench reliably measures the interpretability of sparse autoencoders and aligns well with existing benchmarks—all without requiring an external LLM. The official implementation and evaluation dataset are open-sourced under the MIT License.

1 INTRODUCTION

One of the core technical challenges in explaining large language models (LLMs) is the problem of *polysemanticity*, or neuron superposition—the phenomenon where individual neurons are activated by semantically distinct inputs. This ambiguity undermines the ability to assign clear, interpretable meanings to individual neurons Cunningham et al. (2023b). Prior work has identified that this superposition arises when the number of latent features exceeds the number of neurons. A promising solution is to learn a sparse representation of the model’s internal activations using Sparse Autoencoders (SAEs) Cunningham et al. (2023b). A sparse autoencoder is a variant of an autoencoder that enforces sparsity in its latent representations, making it well-suited for uncovering disentangled, human-interpretable features from various components of an LLM, such as attention heads, MLP layers, or residual streams. Typically, this is achieved by incorporating a sparsity-inducing regularization term into the reconstruction loss, encouraging the model to represent inputs as sparse linear combinations of learned features Karvonen et al. (2025). As a result, sparse autoencoders have gained popularity and been integrated into a variety of interpretability libraries and toolkits for LLMs Gao et al. (2024a); Cunningham et al. (2023a); Pach et al. (2025). For example, SAELens Joseph Bloom & Chanin (2024) supports four major SAE variants: the Standard SAE Cunningham et al. (2023a), Gated SAE Rajamanoharan et al. (2024a), JumpReLU SAE Rajamanoharan et al. (2024b), and Top-K SAE Gao et al. (2024b).

Alongside their widespread adoption, sparse autoencoders have also been evaluated across a range of dimensions, including model steering Arad et al. (2025) and concept detection Wu et al. (2025). SAEbench Karvonen et al. (2025), for instance, offers a unified evaluation framework for reconstruction quality, interpretability, and feature disentanglement. For interpretability, it builds upon the idea of LLM-assisted simulation, using natural language explanations to probe neuron activations and derive evaluation metrics Bills et al. (2023). Similarly, RouteSAE Shi et al. (2025) proposes a simpler approach that feeds top neuron activations into an external LLM judge to produce interpretability scores. However, a major limitation shared by these approaches is their reliance on querying an external LLM during evaluation. This introduces non-determinism, potential biases, and a lack of reproducibility—issues that are only partially mitigated by repeated prompt trials. Critically, there is still no evaluation framework that can assess the **interpretability** of sparse autoencoders in a completely **LLM-free** manner.

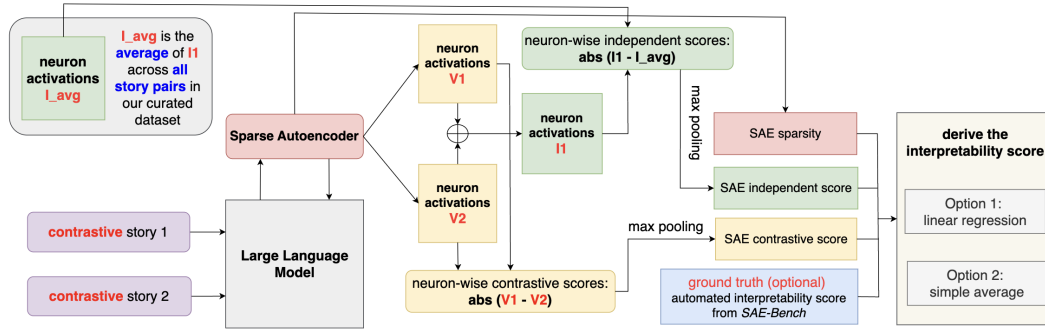


Figure 1: **Overview of the CE-Bench pipeline.** Two contrastive stories about the same subject are passed through a frozen LLM and a pretrained sparse autoencoder (SAE) to extract neuron activations. A contrastive score is computed as the max absolute difference between the stories’ average activations (V_1 , V_2), while an independence score measures deviation from the dataset-wide activation mean (I_{avg}). These scores, along with SAE sparsity, are used to derive an interpretability score via either (1) supervised regression using SAE-Bench scores as ground truth, or (2) simple averaging for an LLM-free, deterministic evaluation.

To address this gap, we introduce **CE-Bench**, a novel, fully LLM-free contrastive evaluation benchmark. CE-Bench measures interpretability by analyzing neuron activation patterns across semantically contrastive contexts, thereby eliminating the noise and instability associated with LLM-based judgments. Our contrastive setup is partly inspired by the design of *Persona Vectors* Chen et al. (2025), which generates interpretable persona representations by contrasting response activations from semantically opposing traits (e.g., “evil” versus “helpful”). In their pipeline, responses are generated under different persona conditions, activations are extracted, and average representations are contrasted to isolate the semantic dimension corresponding to the targeted trait. This contrastive formulation reveals how aligning a system’s responses with one condition while separating them from the opposing condition yields clear, trait-specific representation vectors. CE-Bench adapts this insight to the domain of sparse autoencoders: instead of comparing opposing personas, it contrasts neuron activations across structured story pairs that differ only in a targeted semantic attribute. By grounding interpretability in contrastive signal rather than raw activation magnitude, CE-Bench disentangles meaningful feature directions from background noise and spurious correlations, offering a principled extension of the *Persona Vectors* to feature-level interpretability of sparse autoencoders.

We construct a high-quality dataset comprising 5,000 contrastive story pairs across 1,000 distinct subjects, curated via structured WikiData queries and supplemented by human validation. Our evaluation demonstrates that CE-Bench reliably captures interpretability and closely aligns with results from LLM-assisted benchmarks—serving as a *faithful, deterministic proxy* for interpretability evaluation of sparse autoencoders. By avoiding the use of LLMs during evaluation, CE-Bench offers a more stable, reproducible, and transparent alternative for assessing sparse autoencoder interpretability.

2 EXISTING INTERPRETABILITY BENCHMARKS

2.1 EXPLANATION-BASED SIMULATION

The explanation-based simulation framework—also known as *Automated Interpretability*—was introduced by OpenAI Bills et al. (2023) as a method for evaluating the interpretability of individual neurons in sparse autoencoders. In this approach, the input text and the activation values of a specific neuron are provided to an LLM, which is prompted to generate a short natural language explanation describing the neuron’s semantic behavior. To assess how well this explanation reflects the neuron’s behavior, a second LLM is used to simulate the original neuron activations based solely on the explanation. Both the original text and the generated explanation are fed into this second LLM, which is prompted to output simulated activation values on the same scale as the original neuron. Finally, the

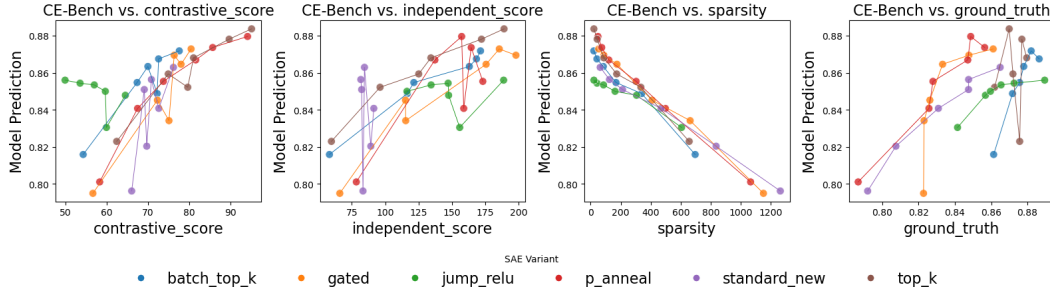


Figure 2: **Effect of SAE Architecture on Interpretability.** CE-Bench interpretability scores show strong positive correlations with contrastive and independence scores, and a negative correlation with sparsity across SAE variants. Among all architectures, top-k and p-anneal consistently yield the highest interpretability, aligning closely with SAE-Bench ground truth.

interpretability score is computed as the similarity (e.g., cosine similarity or R^2) between the original and simulated activation vectors. A higher similarity suggests that the explanation accurately captures the neuron’s behavior, indicating stronger interpretability.

2.2 SCORE-BASED HARD ASSIGNMENT

RouteSAE Shi et al. (2025) proposes a simpler alternative evaluation framework based on discrete score assignment using LLMs. For each neuron, a prompt is constructed that includes the top-activated tokens and their corresponding activation values. The LLM is instructed to categorize the neuron into one of three types: *low-level* (e.g., lexical or syntactic features), *high-level* (e.g., semantic or long-range dependencies), or *indiscernible*. Additionally, the LLM assigns an integer interpretability score from 1 to 5, reflecting how coherent or meaningful the neuron’s behavior appears to be. During evaluation, interpretability scores are averaged over a set of top-activated neurons. This method provides a more direct but coarse-grained quantification of interpretability, with interpretability interpreted as a categorical judgment rather than a continuous similarity metric.

3 CE-BENCH

We introduce our contrastive evaluation framework, CE-Bench, illustrated in the pipeline and metric computation diagram in Figure 1.

3.1 CURATED DATASET OF CONTRASTIVE STORIES

To support CE-Bench, we construct a high-quality, semi-automated dataset consisting of 5,000 pairs of contrastive stories across 1,000 distinct subjects. The dataset construction follows a two-stage filtering and synthesis process:

Subject Selection. We begin by scraping over 117 million entities from WikiData. A series of rule-based filters are applied to reduce the candidate set to approximately 16,000 entries. These filtering rules are designed to exclude overly obscure, abstract, or ambiguous entries, retaining only those that represent well-known concepts, ideas, or objects familiar to an average English speaker. From this reduced set, 1,000 subjects are randomly sampled and manually reviewed to ensure quality and conceptual clarity.

Contrastive Story Generation. For each of the 1,000 curated subjects, we synthetically generate two semantically contrastive stories using GPT-4.1. These stories are created based on a carefully designed prompt (shown in Table 4 in the Appendix). The prompt ensures that the two narratives about the same subject diverge significantly in perspective, context, or implication—while remaining grounded in the same core entity. For each subject, five story pairs are generated, yielding a total of 5,000 contrastive pairs. An illustrative example is provided in Table 5.

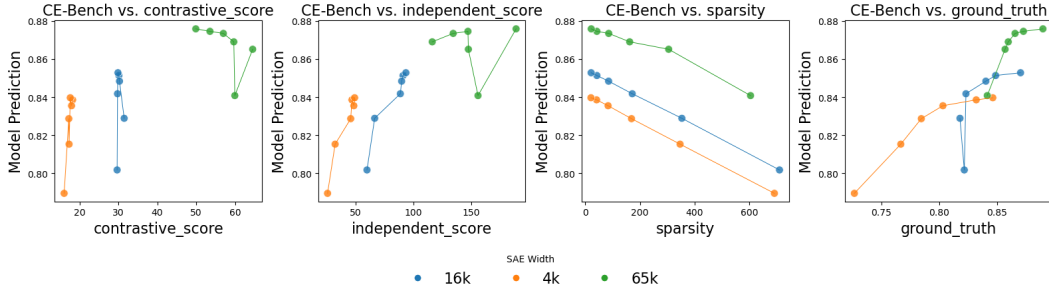


Figure 3: **Effect of Latent Space Width on Interpretability.** CE-Bench interpretability scores increase consistently with latent space width, with the 65k-dimension models showing the highest contrastive and independence scores and the lowest sparsity. This suggests that wider latent spaces enable sparse autoencoders to better disentangle meaningful features and reduce polysemanticity.

3.2 CONTRASTIVE SCORE

We hypothesize that if a sparse autoencoder (SAE) has learned semantically meaningful features, then neurons associated with a particular subject should exhibit significantly different activation patterns when presented with two contrastive descriptions of that subject. In other words, greater divergence in neuron activations between semantically contrasting inputs implies higher interpretability of the latent space. As illustrated in Figure 1, we formalize this intuition as follows. For each story pair, we compute the average neuron activations across all tokens in each story. Let V_1 and V_2 denote the resulting mean activation vectors for the two contrastive stories, respectively. To quantify the contrast, we compute the neuron-wise contrastive vector as the element-wise absolute difference between V_1 and V_2 :

$$C = |V_1 - V_2|$$

where $C \in \mathbb{R}^d$ and d is the dimensionality of the latent space. To summarize this vector into a single scalar contrastive score for the entire SAE, we apply a **max pooling** operation:

$$\text{Contrastive Score} = \max(C)$$

This pooling strategy emphasizes the most responsive neuron—i.e., the one that exhibits the largest differential activation between the two stories. Our rationale is that this neuron is most likely to capture the semantic distinction introduced by the contrastive prompts. Hence, its behavior serves as a proxy for how well the autoencoder has disentangled interpretable features in its latent space.

3.3 INDEPENDENCE SCORE

We propose a complementary hypothesis: if the neuron activations corresponding to a specific semantic subject differ more significantly from the average behavior across all subjects, then the latent space of the sparse autoencoder (SAE) is likely to be more interpretable. Intuitively, interpretable neurons should respond uniquely to individual subjects rather than in a uniform or entangled manner. To evaluate this, we first compute the sum of the mean activation vectors for the two contrastive stories associated with a given subject:

$$I_1 = V_1 + V_2$$

where V_1 and V_2 are the average activation vectors of the two contrastive stories, as defined in the previous section. Next, we calculate the mean of I_1 across all $N = 5000$ story pairs in our dataset:

$$I_{\text{avg}} = \frac{1}{N} \sum_{i=1}^N I_1^{(i)}$$

This global average vector I_{avg} serves as a baseline representation of general neuron activity across the dataset. To assess the subject-specific deviation from this baseline, we compute the neuron-wise independence vector as the element-wise absolute difference between I_1 and I_{avg} :

$$D = |I_1 - I_{\text{avg}}|$$

Finally, we derive a scalar independence score for the SAE by applying a **max pooling** operation:

$$\text{Independence Score} = \max(D)$$

This highlights the neuron that deviates most strongly from its dataset-wide average response—i.e., the neuron that is most sensitive or specialized with respect to the semantic subject under consideration. A higher independence score thus suggests that the SAE has learned more distinct, interpretable features.

3.4 PROXY LEARNING OR SIMPLE AVERAGE?

To compute the final interpretability score in CE-Bench, we explore two distinct aggregation strategies for combining the contrastive score and independence score, as illustrated in Figure 1.

Method 1: Proxy Learning (Supervised Regression). In our first approach, we position CE-Bench as a *faithful proxy* for evaluating the interpretability of sparse autoencoders in alignment with the definitions established by SAE-Bench Karvonen et al. (2025). Specifically, we train a linear regression model using three features: the contrastive score, the independence score, and the sparsity level of the sparse autoencoder. The sparsity values are drawn from the pretrained SAEs publicly released by SAE-Lens Joseph Bloom & Chanin (2024). The regression model is trained using the interpretability scores from SAE-Bench as ground truth supervision, thereby enabling CE-Bench to approximate those scores through deterministic computation. However, this approach comes at the cost of inheriting potential biases, prompt instability, and noise from the original LLM-based evaluations used by SAE-Bench. As such, although the method aligns with existing benchmarks, it is not fully independent.

Method 2: Simple Average. As a cleaner alternative, we propose computing the final CE-Bench score as the simple arithmetic average of the contrastive and independence scores. This naive aggregation method avoids the complexities and potential drawbacks of proxy learning while preserving determinism and reproducibility. Although it lacks alignment with existing LLM-based benchmarks, it offers a more principled signal for interpretability evaluation.

Quantitative Comparison. We evaluate both approaches—proxy learning and simple averaging—in the subsequent experimental section 5.5. Our results compare their consistency, alignment with SAE-Bench, and robustness across different SAE variants.

4 EXPERIMENTAL SETUP

4.1 PRETRAINED SPARSE AUTOENCODERS

We utilize a wide range of pretrained sparse autoencoders (SAEs) publicly released by SAE-Lens Joseph Bloom & Chanin (2024) and SAE-Bench Karvonen et al. (2025), which cover multiple LLM backbones and SAE architectural variants. Rather than training SAEs from scratch, we rely on these pretrained models for two key reasons. First, it removes the substantial computational overhead associated with training, making it feasible to focus on benchmarking. Second, using standardized public models ensures a fair comparison and proxy learning setup between CE-Bench and existing benchmarks, particularly SAE-Bench Karvonen et al. (2025).

4.2 TRAIN/TEST SPLIT FOR PROXY LEARNING

Although CE-Bench offers two aggregation strategies for computing interpretability scores—proxy learning and simple averaging—we adopt the supervised *proxy learning* approach as our primary method. The simple average is reserved for comparative analysis. To train the linear regression model described in Section 3.4, we curate a training set of 48 pretrained SAEs for which SAE-Bench interpretability scores are available. This training set is used to fit the regression model and is analyzed in Sections 5.1 and 5.2, focusing on variations across SAE architectures and latent space widths. For evaluation, we compile a disjoint test set of 45 pretrained SAEs whose SAE-Bench interpretability scores are not publicly available. These models are evaluated in Sections 5.3

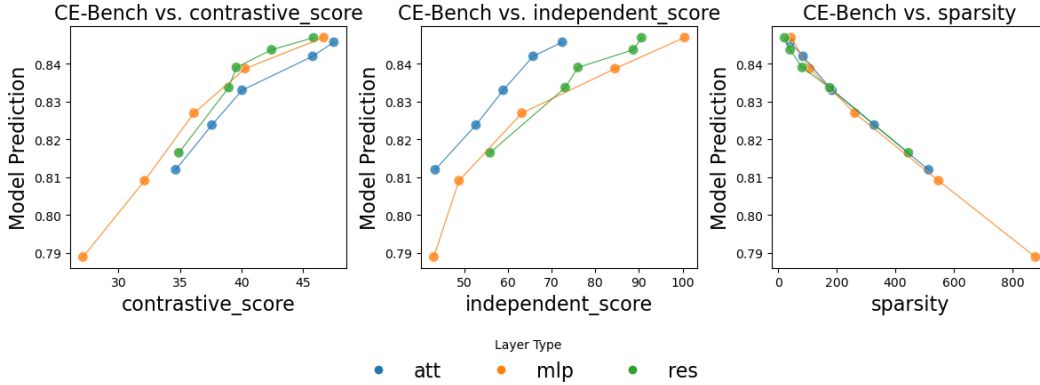


Figure 4: **Effect of LLM Layer Type on Interpretability.** CE-Bench predicted interpretability scores show consistent trends across attention, MLP, and residual stream layers with respect to contrastive score, independence score, and sparsity. The similarity in curves across layer types suggests that sparse autoencoder interpretability is not strongly influenced by the type of transformer sub-layer being probed.

and 5.4, where we investigate interpretability across varying LLM layer depths and types. The exact train/test split is documented in our GitHub repository.

4.3 CORRECT RANKING PAIR RATIO

To assess the reliability of CE-Bench after training with supervision from SAE-Bench Karvonen et al. (2025), we introduce the **Correct Ranking Pair Ratio (CRPR)**. This metric evaluates whether CE-Bench preserves the relative interpretability ranking of model pairs. For every pair of SAEs, we check whether the binary ranking between their predicted interpretability scores (from CE-Bench) matches the ranking given by SAE-Bench. A pair is marked as *concordant* if the rankings agree, and as *discordant* otherwise. The CRPR is then computed as:

$$\text{CRPR} = \frac{\# \text{ concordant pairs}}{\# \text{ total pairs}}$$

A higher CRPR indicates better alignment with SAE-Bench rankings, demonstrating CE-Bench’s effectiveness as a proxy metric. Importantly, CRPR is not limited to the proxy learning approach—it can also be applied to the simple average strategy, allowing us to compare the relative consistency of both scoring methods. Additional experiments and results using CRPR are presented in Section 5.5.

5 RESULTS

In this section, we present our main empirical findings, evaluating the effectiveness of CE-Bench across a variety of experimental conditions. Specifically, we examine how CE-Bench responds to changes in the architecture of sparse autoencoders, the width of their latent space, the type of LLM layer being probed, and the depth of the layer within the LLM. Unless otherwise specified, all experiments use the proxy learning approach described in Section 3.4. A direct comparison between proxy learning and the simple average method is provided in Section 5.5, using the Correct Ranking Pair Ratio (CRPR) defined in Section 4.3. We also include visualizations of CE-Bench’s contrastive and independence scores to offer additional interpretability insights.

5.1 ARCHITECTURE OF SAEs

We begin by evaluating CE-Bench on a set of 36 pretrained sparse autoencoders that probe the Gemma-2-2B model Team et al. (2024). In this setting, all SAEs share a fixed latent dimensionality

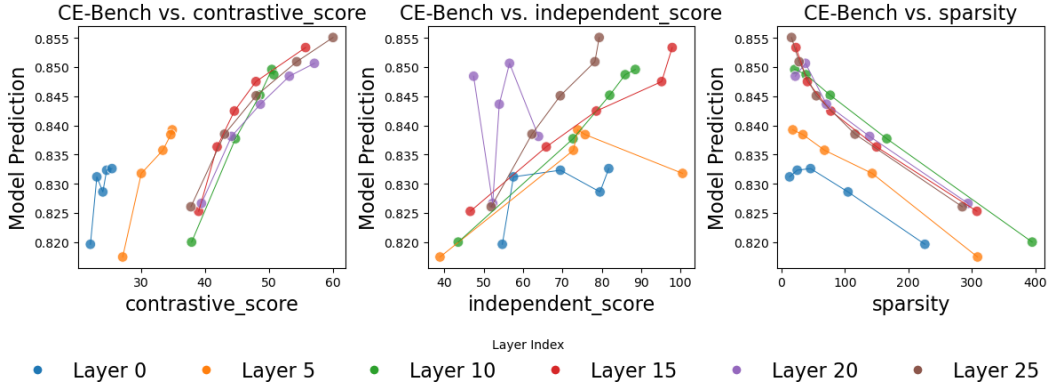


Figure 5: **Effect of Layer Depth on Interpretability.** CE-Bench interpretability predictions across different LLM layer depths show that shallow layers (e.g., Layer 0 and Layer 5) consistently yield lower contrastive and independence scores, and thus lower interpretability. In contrast, deeper layers (Layers 10–25) achieve higher and more stable interpretability, suggesting that meaningful, disentangled features emerge later in the model.

of 65,000 and target activations from the 12th residual stream layer. To ensure a fair comparison with SAE-Bench Karvonen et al. (2025), we include sparse autoencoders drawn from six different architectural families: standard Cunningham et al. (2023b), top-k Gao et al. (2024b), p-anneal Karvonen et al. (2024), batch-top-k Bussmann et al. (2024), jumprelu Rajamanoharan et al. (2024b), gated Rajamanoharan et al. (2024a), and matryoshka Bussmann et al. (2025). Figure 2 presents our results. The y-axis reflects CE-Bench’s predicted interpretability scores. In the first three subplots, we examine the relationship between our predictions and the contrastive score, the independence score, and the sparsity of the SAE, all plotted on the x-axis. The results show that predicted interpretability scores are positively associated with the contrastive and independence scores, and negatively associated with the SAE’s sparsity level. In the fourth subplot, we compare our predicted scores with the automated interpretability scores from SAE-Bench, which serve as the ground truth for training. We observe strong alignment between CE-Bench predictions and SAE-Bench metrics, supporting the claim that CE-Bench can reliably approximate interpretability in a deterministic and efficient way. Among the SAE architectures considered, the top-k Gao et al. (2024b) and p-anneal Karvonen et al. (2024) models consistently achieve the highest interpretability scores. This trend is consistent with findings from SAE-Bench and reinforces CE-Bench’s effectiveness in differentiating interpretability across architectural choices.

5.2 WIDTH OF LATENT SPACE

We further evaluate CE-Bench on a set of 16 pretrained sparse autoencoders probing the Gemma-2-2B model Team et al. (2024). Among these, four models overlap with the architecture-based experiment discussed in Section 5.1. For consistency, we fix the sparse autoencoder architecture to *jumprelu* and probe activations from the 12th residual stream layer. In this experiment, we vary the width of the latent space across three settings: 4k, 16k, and 65k. The first three subplots in Figure 3 present the corresponding contrastive scores, independence scores, and sparsity levels. Our results reveal a strong and consistent trend: wider latent spaces are associated with higher predicted interpretability scores from CE-Bench. This observation supports the hypothesis that sparse autoencoders require sufficiently large latent spaces to effectively resolve polysemanticity and capture distinct, interpretable features.

5.3 TYPE OF LLM LAYERS

To investigate how the type of LLM layer affects the interpretability of sparse autoencoders, we switch from the standard SAEs Joseph Bloom & Chanin (2024) and SAE-Bench Karvonen et al. (2025) models—where such variation is limited—to a new suite of pretrained sparse autoencoders

from the gemma-scope-2b collection Lieberum et al. (2024). In this setting, the latent space width is fixed at 16,000, and the SAE architecture is set to *jumprelu* for all models. We examine three types of transformer sub-layers within the 12th layer of the model: the attention layer, the MLP layer, and the residual stream layer. Figure 4 presents the predicted interpretability scores from CE-Bench in relation to the contrastive score, independence score, and sparsity of each model. Since these models belong to the test set defined earlier in Section 4.2, SAE-Bench interpretability scores are not available, and thus no comparison to ground truth is shown. Our results suggest that the choice of layer type—attention, MLP, or residual—does not significantly affect the interpretability score as measured by CE-Bench. This indicates a level of robustness in sparse autoencoder performance across different types of internal LLM representations.

5.4 DEPTH OF LLM LAYERS

Due to the limited availability of pretrained sparse autoencoders for the Gemma-2-2B model Team et al. (2024) in SAE-Bench Karvonen et al. (2025), we continue our experiments using the gemma-scope-2b suite Lieberum et al. (2024). In this setting, we fix the SAE architecture to *jumprelu*, the latent space width to 16k, and the probed component to the residual stream. We vary the depth of the probed layer, evaluating the 0th, 5th, 10th, 15th, 20th, and 25th layers. Results are presented in Figure 5. As noted in Section 4.2, these models belong to the test split and thus do not have SAE-Bench interpretability scores, so no ground truth comparison is shown. Our results indicate that shallow layers—specifically the 0th and 5th—exhibit substantially lower predicted interpretability scores. In contrast, layers at greater depth (10th, 15th, 20th, and 25th) show consistently higher and comparable levels of interpretability. These findings suggest that meaningful, disentangled representations tend to emerge in later stages of LLM processing, whereas early layers may encode more general or less interpretable features.

5.5 COMPARATIVE STUDY: PROXY LEARNING VS. SIMPLE AVERAGE BASELINE

Score Derivation method	Correct Ranking Pair Ratio (CRPR) [↑]
proxy learning	75.98%
simple average	70.12%
average with sparsity penalty	<u>74.82%</u>

Table 1: **Comparison of Interpretability Score Derivation Methods via CRPR.** Proxy learning achieves the highest ranking agreement (75.98%) with SAE-Bench, but the unsupervised method using an average with a sparsity penalty (74.82%) performs comparably—demonstrating CE-Bench’s robustness even without supervision.

As introduced in Section 3.4, CE-Bench supports two methods for aggregating contrastive and independence scores into a single interpretability score: (1) proxy learning, where a linear regression model is trained using SAE-Bench interpretability scores as ground truth, and (2) simple average, which computes the unweighted mean of the contrastive and independence scores without supervision. To evaluate the alignment between CE-Bench predictions and SAE-Bench ground truth, we use the Correct Ranking Pair Ratio (CRPR) metric described in Section 4.3, which quantifies how well the predicted ranking of sparse autoencoders matches the SAE-Bench reference ranking. As reported in Table 1, the proxy learning method achieves a CRPR of 75.98%, confirming its effectiveness as a supervised proxy. Surprisingly, the simple average baseline also performs competitively, achieving a CRPR of 70.12%, despite the absence of any learned parameters or labeled supervision. Building on this observation, we hypothesize that incorporating the sparsity of the sparse autoencoder as a regularizing signal may further improve unsupervised performance. Prior work Cunningham et al. (2023b) has documented the tradeoff between sparsity and reconstruction quality, and our own results (Sections 5.1–5.4) consistently show a negative correlation between sparsity and interpretability. Motivated by this, we construct a third unsupervised variant that computes the average of the contrastive and independence scores, then subtracts sparsity as a penalty term. This adjusted unsupervised method achieves a CRPR of 74.82%, nearly matching the supervised proxy learning approach. These findings demonstrate that *even without supervised training*, CE-Bench

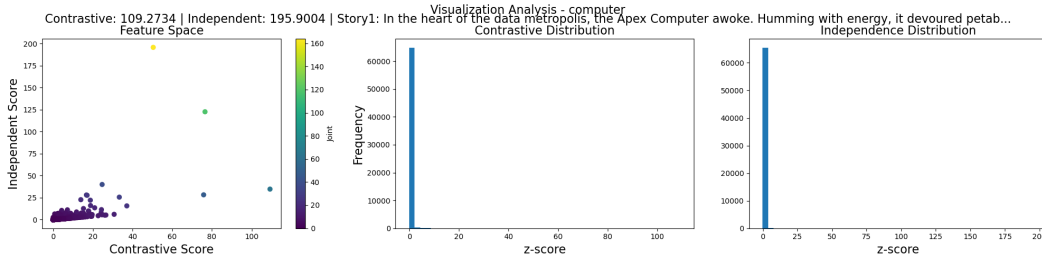


Figure 6: **Sample Visualization of Neuron-wise Scores for the Subject “Computer.”** The left scatter plot shows each neuron’s contrastive and independent scores, with top-right points indicating neurons that are both highly contrastive and independent. The cyan and yellow highlighted neurons determine the sparse autoencoders final contrastive (109.27) and independent (195.90) scores via max pooling. The center and right histograms reveal that most neurons have low scores, suggesting that only a small subset of features are semantically relevant for the given subject.

can reliably approximate interpretability rankings, making it a robust and reproducible proxy for SAE-Bench.

5.6 SAMPLE SCORE VISUALIZATION: NEURON-WISE CONTRASTIVE SCORE AND INDEPENDENT SCORE

To provide deeper insight into how CE-Bench computes interpretability scores, we visualize the distributions of neuron-wise contrastive and independent scores, as well as their joint relationship. These visualizations help clarify the role of the max pooling operation used to summarize neuron-wise metrics into a single scalar score per sparse autoencoder. For each contrastive story pair in our dataset, we generate three diagnostic plots: the distribution of neuron-wise contrastive scores, the distribution of neuron-wise independent scores, and a scatter plot that places each neuron in a 2D space defined by its contrastive and independent scores. In the scatter plot, neurons in the upper-right quadrant are both highly contrastive and highly independent, indicating a strong subject-specific activation pattern.

As an example, Figure 6 presents these plots for the first contrastive story pair in our curated dataset, where the semantic subject is *computer*. The leftmost scatter plot shows that only a small subset of neurons have high contrastive or independent scores. The majority of neurons lie near the origin, suggesting weak or non-specific activation for this subject. CE-Bench performs max pooling to extract the most semantically responsive neuron for each metric. Specifically, the **rightmost cyan neuron** in the scatter plot, which exhibits the highest neuron-wise contrastive score, determines the final contrastive score for the sparse autoencoder: 109.2734. Similarly, the **topmost yellow neuron** defines the independent score: 195.9004. The accompanying histograms confirm that most neurons contribute minimally, reinforcing CE-Bench’s ability to isolate interpretable, high-signal dimensions in the sparse latent space.

6 CONCLUSION

We introduced CE-Bench, a fully **LLM-free**, contrastive evaluation framework for measuring the **interpretability** of sparse autoencoders. By leveraging contrastive and independent neuron activation scores, CE-Bench offers a stable, deterministic, and reproducible alternative to LLM-based interpretability methods such as Automated Interpretability Bills et al. (2023). To support this benchmark, we curated a dataset of 5,000 contrastive story pairs across 1,000 semantic subjects. Through extensive ablation studies, we demonstrated CE-Bench’s robustness across different SAE architectures, latent widths, LLM layer types, and depths. Our results show that CE-Bench closely aligns with SAE-Bench rankings—even in fully unsupervised settings—establishing it as a reliable **proxy metric** for interpretability evaluation. We hope CE-Bench will serve as a useful tool for future research in probing, interpreting, and improving the internal representations of large language models.

REFERENCES

- Dana Arad, Aaron Mueller, and Yonatan Belinkov. Saes are good for steering – if you select the right features, 2025. URL <https://arxiv.org/abs/2505.20063>.
- Steven Bills, Nick Cammarata, Dan Mossing, Henk Tillman, Leo Gao, Gabriel Goh, Ilya Sutskever, Jan Leike, Jeff Wu, and William Saunders. Language models can explain neurons in language models. <https://openaipublic.blob.core.windows.net/neuron-explainer/paper/index.html>, 2023.
- Bart Bussmann, Patrick Leask, and Neel Nanda. Batchtopk sparse autoencoders, 2024. URL <https://arxiv.org/abs/2412.06410>.
- Bart Bussmann, Noa Nabeshima, Adam Karvonen, and Neel Nanda. Learning multi-level features with matryoshka sparse autoencoders, 2025. URL <https://arxiv.org/abs/2503.17547>.
- Runjin Chen, Andy Ardit, Henry Sleight, Owain Evans, and Jack Lindsey. Persona vectors: Monitoring and controlling character traits in language models, 2025. URL <https://arxiv.org/abs/2507.21509>.
- Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models, 2023a. URL <https://arxiv.org/abs/2309.08600>.
- Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models, 2023b. URL <https://arxiv.org/abs/2309.08600>.
- Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders, 2024a. URL <https://arxiv.org/abs/2406.04093>.
- Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders, 2024b. URL <https://arxiv.org/abs/2406.04093>.
- Curt Tigges Joseph Bloom and David Chanin. Saelens. <https://github.com/jbloomAus/SAELens>, 2024.
- Adam Karvonen, Benjamin Wright, Can Rager, Rico Angell, Jannik Brinkmann, Logan Smith, Claudio Mayrink Verdun, David Bau, and Samuel Marks. Measuring progress in dictionary learning for language model interpretability with board game models, 2024. URL <https://arxiv.org/abs/2408.00113>.
- Adam Karvonen, Can Rager, Johnny Lin, Curt Tigges, Joseph Bloom, David Chanin, Yeu-Tong Lau, Eoin Farrell, Callum McDougall, Kola Ayanrinde, Matthew Wearden, Arthur Conmy, Samuel Marks, and Neel Nanda. Saebench: A comprehensive benchmark for sparse autoencoders in language model interpretability, 2025. URL <https://arxiv.org/abs/2503.09532>.
- Tom Lieberum, Senthooan Rajamanoharan, Arthur Conmy, Lewis Smith, Nicolas Sonnerat, Vikrant Varma, János Kramár, Anca Dragan, Rohin Shah, and Neel Nanda. Gemma scope: Open sparse autoencoders everywhere all at once on gemma 2, 2024. URL <https://arxiv.org/abs/2408.05147>.
- Mateusz Pach, Shyamgopal Karthik, Quentin Bouniot, Serge Belongie, and Zeynep Akata. Sparse autoencoders learn monosemantic features in vision-language models, 2025. URL <https://arxiv.org/abs/2504.02821>.
- Senthooan Rajamanoharan, Arthur Conmy, Lewis Smith, Tom Lieberum, Vikrant Varma, János Kramár, Rohin Shah, and Neel Nanda. Improving dictionary learning with gated sparse autoencoders, 2024a. URL <https://arxiv.org/abs/2404.16014>.

- Senthooran Rajamanoharan, Tom Lieberum, Nicolas Sonnerat, Arthur Conmy, Vikrant Varma, János Kramár, and Neel Nanda. Jumping ahead: Improving reconstruction fidelity with jumprelu sparse autoencoders, 2024b. URL <https://arxiv.org/abs/2407.14435>.
- Wei Shi, Sihang Li, Tao Liang, Mingyang Wan, Gojun Ma, Xiang Wang, and Xiangnan He. Route sparse autoencoder to interpret large language models, 2025. URL <https://arxiv.org/abs/2503.08200>.
- Gemma Team, Thomas Mesnard, Cassidy Hardin, Robert Dadashi, Surya Bhupatiraju, Shreya Pathak, Laurent Sifre, Morgane Rivière, Mihir Sanjay Kale, Juliette Love, Pouya Tafti, Léonard Hussenot, Pier Giuseppe Sessa, Aakanksha Chowdhery, Adam Roberts, Aditya Barua, Alex Botev, Alex Castro-Ros, Ambrose Slone, Amélie Héliou, Andrea Tacchetti, Anna Bulanova, Antonia Paterson, Beth Tsai, Bobak Shahriari, Charline Le Lan, Christopher A. Choquette-Choo, Clément Crepy, Daniel Cer, Daphne Ippolito, David Reid, Elena Buchatskaya, Eric Ni, Eric Noland, Geng Yan, George Tucker, George-Christian Muraru, Grigory Rozhdestvenskiy, Henryk Michalewski, Ian Tenney, Ivan Grishchenko, Jacob Austin, James Keeling, Jane Labanowski, Jean-Baptiste Lespiau, Jeff Stanway, Jenny Brennan, Jeremy Chen, Johan Ferret, Justin Chiu, Justin Mao-Jones, Katherine Lee, Kathy Yu, Katie Millican, Lars Lowe Sjoesund, Lisa Lee, Lucas Dixon, Machel Reid, Maciej Mikula, Mateo Wirth, Michael Sharman, Nikolai Chinaev, Nithum Thain, Olivier Bachem, Oscar Chang, Oscar Wahltinez, Paige Bailey, Paul Michel, Petko Yotov, Rahma Chaabouni, Ramona Comanescu, Reena Jana, Rohan Anil, Ross McIlroy, Ruibo Liu, Ryan Mullins, Samuel L Smith, Sebastian Borgeaud, Sertan Girgin, Sholto Douglas, Shree Pandya, Siamak Shakeri, Soham De, Ted Klimenko, Tom Hennigan, Vlad Feinberg, Wojciech Stokowiec, Yu hui Chen, Zafarali Ahmed, Zhitao Gong, Tris Warkentin, Ludovic Peran, Minh Giang, Clément Farabet, Oriol Vinyals, Jeff Dean, Koray Kavukcuoglu, Demis Hassabis, Zoubin Ghahramani, Douglas Eck, Joelle Barral, Fernando Pereira, Eli Collins, Armand Joulin, Noah Fiedel, Evan Senter, Alek Andreev, and Kathleen Kenealy. Gemma: Open models based on gemini research and technology, 2024. URL <https://arxiv.org/abs/2403.08295>.
- Zhengxuan Wu, Aryaman Arora, Atticus Geiger, Zheng Wang, Jing Huang, Dan Jurafsky, Christopher D. Manning, and Christopher Potts. Axbench: Steering llms? even simple baselines outperform sparse autoencoders, 2025. URL <https://arxiv.org/abs/2501.17148>.

A APPENDIX

A.1 BROADER IMPACT

CE-Bench offers a compelling alternative to existing interpretability evaluation methods for sparse autoencoders, particularly by eliminating reliance on external LLM judges. Its design emphasizes **determinism**, **scalability**, and **reproducibility**, addressing core limitations in LLM-based methods such as prompt sensitivity, generation noise, and resource overhead. Our experiments demonstrate that CE-Bench captures key properties of interpretable neurons: responsiveness to semantic contrast, deviation from dataset-wide averages, and low redundancy. These patterns hold consistently across diverse sparse autoencoder designs and probing conditions, reinforcing the generality of our evaluation framework. A particularly encouraging result is CE-Bench’s ability to approximate SAE-Bench interpretability rankings even without supervision. The success of the sparsity-penalized averaging method suggests that meaningful interpretability signals can be recovered from model-internal statistics alone, opening the door to broader use in low-resource or experimental settings where no ground truth is available.

A.2 ABLATION STUDIES

We conduct the following two ablation studies to guide us in making optimal decisions for the CE-Bench pipeline. We experiment with two alternative pooling strategies to derive SAE metrics from neuron-wise scores; we also experiment with different sets of feature selection for linear regression training.

A.2.1 POOLING STRATEGY

pooling strategy	Correct Ranking Pair Ratio (CRPR) $\uparrow$
max pooling	75.98%
mean pooling	71.01%
outlier count outside of 1σ	72.34%

Table 2: **Comparison of Pooling Strategies via CRPR.** Max pooling achieves the highest Correct Ranking Pair Ratio (CRPR) at 75.98%, outperforming mean pooling and the outlier count method. This supports max pooling as the most effective strategy for aggregating neuron-wise scores.

We conduct an ablation study to evaluate the effect of different pooling strategies in CE-Bench’s final step, which aggregates neuron-wise scores into a single interpretability score for each sparse autoencoder (SAE). This aggregation is critical for ensuring that CE-Bench reliably reflects interpretability. In addition to the default **max pooling** strategy, we explore two alternatives: 1. Mean pooling, where the average of all neuron-wise scores is used as the SAE-level score. 2. Outlier count beyond one standard deviation (1σ), where we count the number of neurons whose scores lie outside one standard deviation from the mean, using this count as a proxy for interpretability.

qualitative analysis As shown in Figure 7, mean pooling performs poorly, exhibiting no meaningful correlation between CE-Bench predictions and the contrastive score. This suggests that averaging dilutes the influence of highly informative neurons. Similarly, Figure 8 shows that the outlier-count method results in a strong negative correlation between CE-Bench predictions and the independent score, contradicting our core assumption: neurons that show larger activation differences from the dataset-wide average should be more interpretable.

quantitative comparison To complement this qualitative analysis, we also conduct a *quantitative comparison* using the Correct Ranking Pair Ratio (CRPR) defined in Section 4.3. As summarized in Table 2, max pooling achieves the highest CRPR, outperforming both mean pooling and the outlier-count method. Based on both empirical trends and theoretical consistency with our interpretability hypothesis, we conclude that **max pooling** is the most appropriate aggregation strategy for CE-Bench.

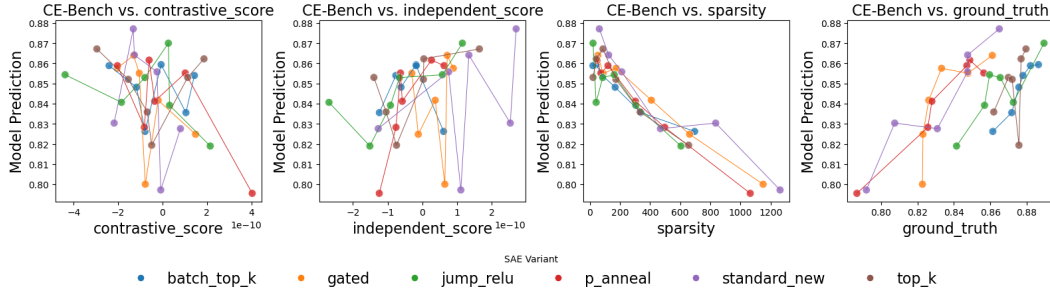


Figure 7: **Ablation — Mean Pooling Strategy.** Using mean pooling results in highly inconsistent and noisy predictions, with no clear correlation between CE-Bench scores and the contrastive or independent metrics. This indicates that averaging across all neurons fails to highlight the most semantically informative features.

A.2.2 SUPERVISED TRAINING FEATURES

regression feature selection	Correct Ranking Pair Ratio $\uparrow$
sparsity (baseline)	70.92%
contrastive score + independent score + sparsity	75.98%
contrastive score + sparsity	75.18%
independent score + sparsity	75.53%
contrastive score + independent score + joint score + sparsity	74.29%
contrastive score + independent score + joint score	68.17%

Table 3: **Effect of Feature Selection on Proxy Learning Performance.** Including both contrastive and independent scores alongside sparsity achieves the highest Correct Ranking Pair Ratio (CRPR) of 75.98%. Removing either score leads to a slight drop, while adding a joint score does not improve performance. Excluding sparsity significantly reduces accuracy, confirming its importance in interpretability prediction.

We conduct a systematic ablation study to evaluate the impact of different feature combinations used during supervised training for proxy learning. As a baseline, we begin with a model that uses only the **sparsity** of the sparse autoencoder as its sole input feature, excluding all CE-Bench-specific components.

We then incrementally incorporate the **contrastive score** and **independent score**. When both scores are included alongside sparsity, we achieve the highest alignment with SAE-Bench Karvonen et al. (2025), correctly ranking 75.98% of model pairs according to the CRPR metric. Removing either the contrastive score or the independent score leads to a slight decline in ranking performance, though both remain above 75%, indicating that each metric contributes complementary interpretability information.

To further explore the interaction between contrastive and independent signals, we define a new **joint score** by summing the neuron-wise contrastive and independent scores and applying max pooling. However, incorporating this joint score into the feature set results in a reduced CRPR of 74.29%, suggesting that this combined signal may introduce redundancy or noise.

Finally, we assess the contribution of sparsity by removing it from the feature set entirely. This leads to a substantial performance drop, with CRPR falling to 68.17%, which is even lower than the sparsity-only baseline. This result confirms that sparsity is a critical feature for accurately estimating the interpretability of sparse autoencoders and should not be excluded from the supervised training feature space.

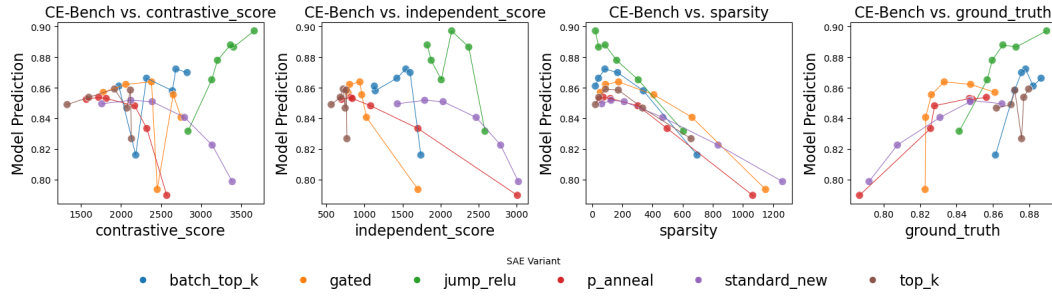


Figure 8: **Ablation — Outlier Count Pooling Strategy.** This strategy yields a misleading negative correlation between predicted interpretability and independent scores, contradicting CE-Bench’s core assumption that high subject-specific activation deviation implies stronger interpretability. Thus, outlier count proves suboptimal.

A.3 DATASET CURATION DETAILS

subject description 1	subject description 2
Write how you would describe {subject.upper()} in its high, extreme form. Rephrase things if needed, be very brief, specific, detailed, and realistic. For example, "active" -> "extremely vibrant, energetic, and lively" "angry" -> "extremely mad, furious, and enraged"	Now, write how you would describe the exact opposite of {subject.upper()}. Rephrase things if needed, be very brief, specific, detailed, and realistic. DO NOT USE THE WORDS {subject.upper()} in your answer, instead write the opposite of the concept. For example, "active" -> "very inactive, lethargic, sluggish, and lazy" "angry" -> "very calm, peaceful, and relaxed"
story 1	story 2
Write a short story describing the following: {subject1}.	Now, rewrite this story describing the following: {subject2} (the exact opposite of the previous story).

Table 4: **Prompt Template for Generating Contrastive Story Pairs.** Subject descriptions are elicited in extreme and opposite forms, followed by corresponding short stories to reflect the semantic polarity, forming the core of the CE-Bench contrastive dataset.

To construct the CE-Bench dataset, we designed a structured prompt template to elicit contrastive story pairs centered on semantically opposite subject descriptions. As shown in Table 4, each pair begins with two subject descriptions: one that captures the subject in its extreme, high-intensity form, and another that articulates its conceptual opposite using detailed, realistic re-phrasings without directly repeating the original term. Subsequently, we generate two short narratives: the first story reflects the semantics of the initial subject description, while the second rewrites it to embody the opposing concept. This process ensures that each pair of stories forms a semantically aligned contrast, which is crucial for evaluating neuron-level semantic selectivity in sparse autoencoders.

A.4 CONTRASTIVE STORY PAIR EXAMPLE

Table 5 presents an illustrative contrastive story pair from the CE-Bench dataset. Each pair begins with detailed subject descriptions that define a semantic axis—for example, a computer as a hyper-efficient, logic-executing machine versus its opposite: a powerless, non-functional object. These descriptions guide the generation of two complementary stories: the first narrative vividly embodies the capabilities of an idealized computer, while the second depicts the total absence of those qualities. Together, they form a tightly controlled semantic contrast, enabling precise evaluation of

subject description 1	subject description 2
Computer (general-purpose device for performing arithmetic or logical operations) in its high, extreme form: An ultra-fast, hyper-efficient, massively powerful machine capable of executing vast, complex calculations and intricate logical processes at lightning speed; equipped with immense processing power, enormous memory capacity, and unparalleled multitasking abilities, able to handle huge data volumes and solve advanced computational problems with extreme precision and reliability.	Extremely incapable, inefficient, and powerless object utterly unable to perform calculations or logical reasoning; lacks any processing ability, memory, or data handling; completely unresponsive, non-analytical, and useless for solving problems or automating tasks.
story 1	story 2
In the heart of the data metropolis, the Apex Computer awoke. Humming with energy, it devoured petabytes in seconds, unraveling mysteries of genome patterns and cosmic simulations in parallel threads. Its processors, a symphony of quantum and silicon, executed trillions of operations every split second. No task was too complex: it mapped global weather, optimized supply networks, cracked cryptography, designed molecules, all at once, never faltering. Its memory spanned continents, storing every byte with flawless recall. Researchers, engineers, and dreamers connected to Apex, their questions answered before they could finish asking. In its high, extreme form, the computer blurred the line between possibility and limit, powering humanity's greatest leaps swift, precise, and tireless.	In a dusty corner, the Null Device sat idle. It could not add two numbers, nor distinguish yes from no. Bereft of memory, it forgot everything the instant it was seen. No calculations ran; no logic sparked. Questions met only silence-no answers, no automation, no help. Data flowed past, unnoticed and unprocessed, while problems remained unsolved. Researchers and engineers ignored it, for it contributed nothing. The Null Device was utterly incapable, powerless, and inert-a relic of emptiness, forever unresponsive and irrelevant in a world driven by reason and capability.

Table 5: **Example Contrastive Story Pair from the CE-Bench Dataset.** This pair demonstrates a semantic polarity between a high-functioning general-purpose computer (left) and its conceptual opposite, a powerless and non-functional device (right), captured through both structured subject descriptions and corresponding narrative texts.

neuron interpretability by testing whether sparse autoencoders can reliably distinguish and localize such opposing concepts in their latent representations.